Naive Bayes & LDA — Engineering Report

Project 008 · Dual-Dataset Classification with Gaussian Naive Bayes & Linear Discriminant Analysis

Project Overview

This project explores **Naive Bayes classification** and **Linear Discriminant Analysis (LDA)** for dimensionality reduction across two semiconductor/IT manufacturing datasets. GaussianNB is trained on each dataset independently with StandardScaler preprocessing, and LDA vs PCA comparisons demonstrate supervised vs unsupervised dimensionality reduction.

Datasets

Dataset	Rows	Features	Target	Classes
Jira Bug Routing	8,000	6 numeric	category	6 (Hardware, Firmware, Software, Performance, Security, Docs)
Silicon Defect Classification	6,000	5 numeric	defect_type	7 (6 defect types + Clean Die)

Methodology

- **Preprocessing:** StandardScaler normalization of all features
- **Model:** GaussianNB — assumes features follow a normal distribution per class
- **Dimensionality Reduction:** LDA (supervised) vs PCA (unsupervised) comparison
- **Evaluation:** Accuracy, precision, recall, F1, confusion matrices
- **Serialization:** joblib pipeline export (gnb_jira.pkl, gnb_silicon.pkl)

Results

Jira Accuracy: 88.6%

Silicon Accuracy: 99.9%

Jira Bug Routing — Per-Class Performance

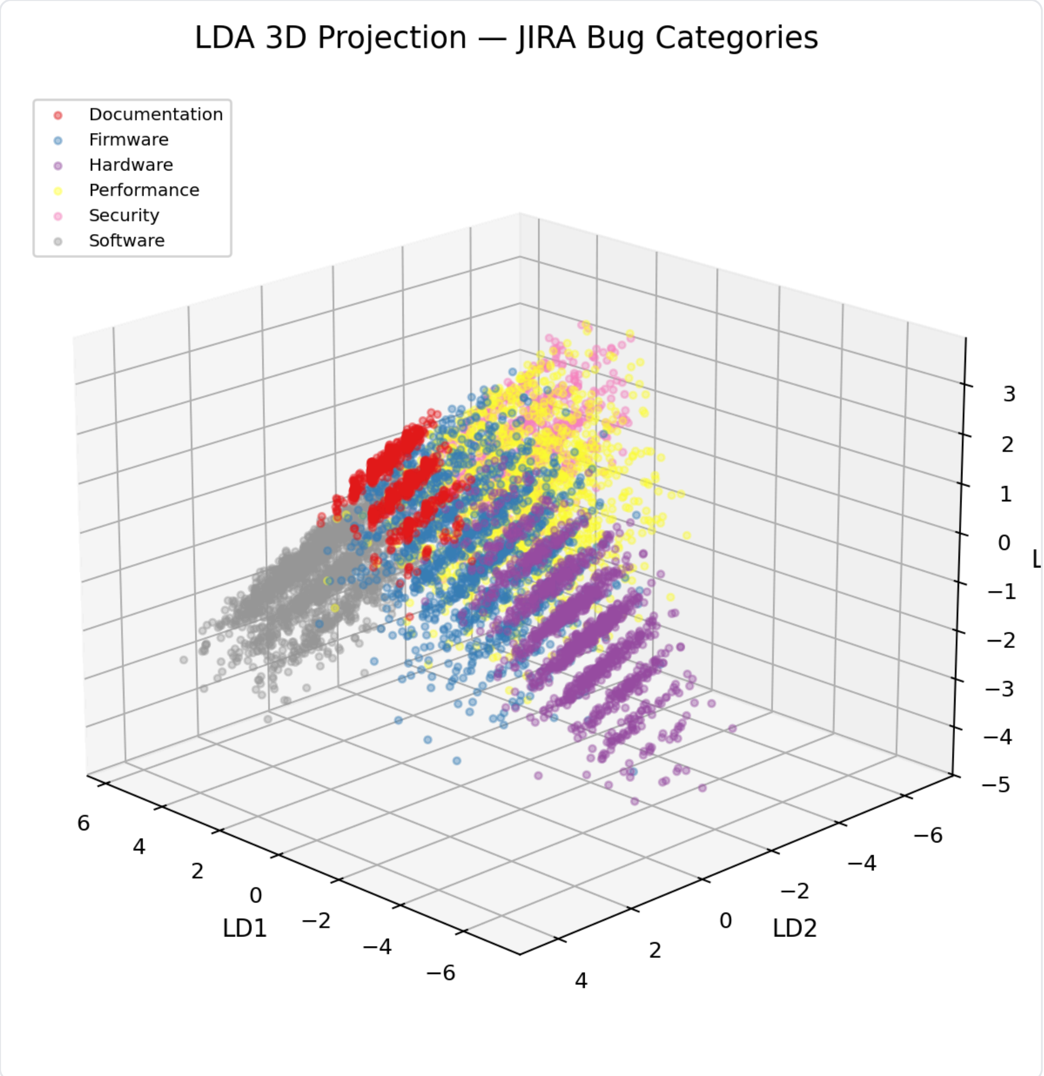
Class	Precision	Recall	F1
Documentation	0.98	0.96	0.97
Firmware	0.73	0.74	0.74
Hardware	0.94	0.94	0.94

Performance	0.79	0.72	0.76
Security	0.88	0.93	0.90
Software	0.94	0.96	0.95

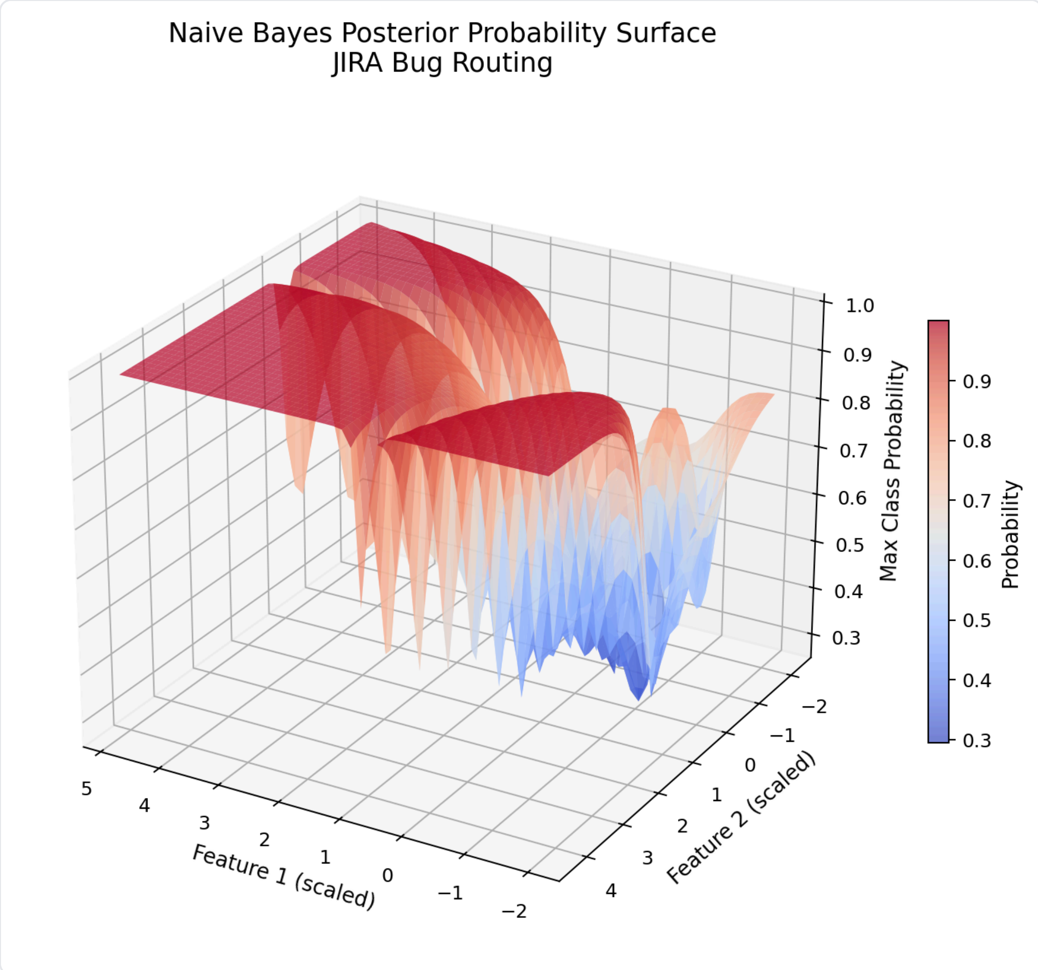
Silicon Defect Classification — Per-Class Performance

Class	Precision	Recall	F1
Clean Die	1.00	1.00	1.00
Gate Oxide Trap	0.99	1.00	1.00
Metal Short	1.00	1.00	1.00
Oxide Pinhole	1.00	0.99	1.00
Particle Defect	1.00	1.00	1.00
Via Open	1.00	1.00	1.00

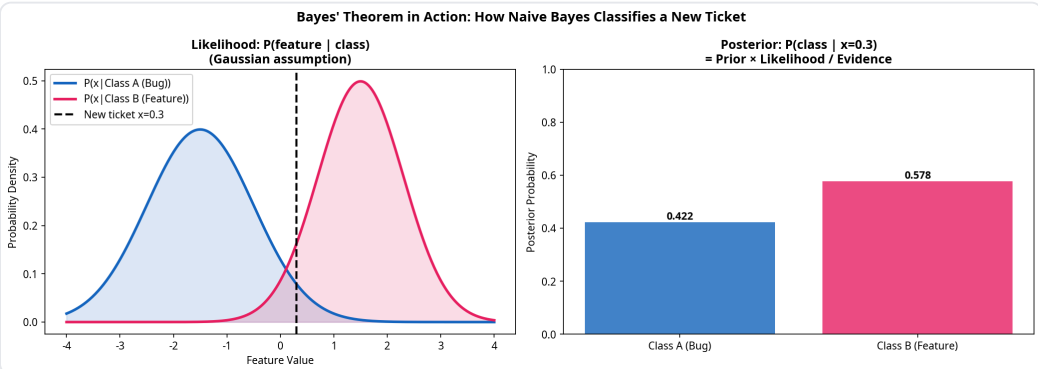
Project 1 — Jira Bug Routing Visualizations



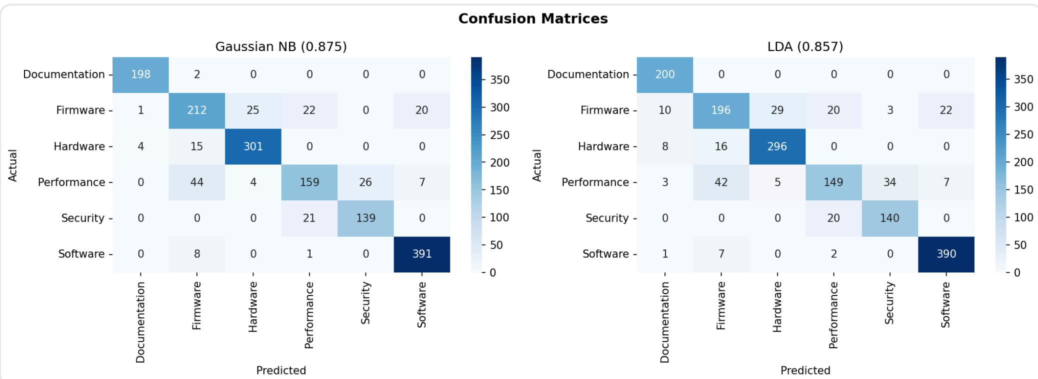
Proj1 Jira 3D Lda



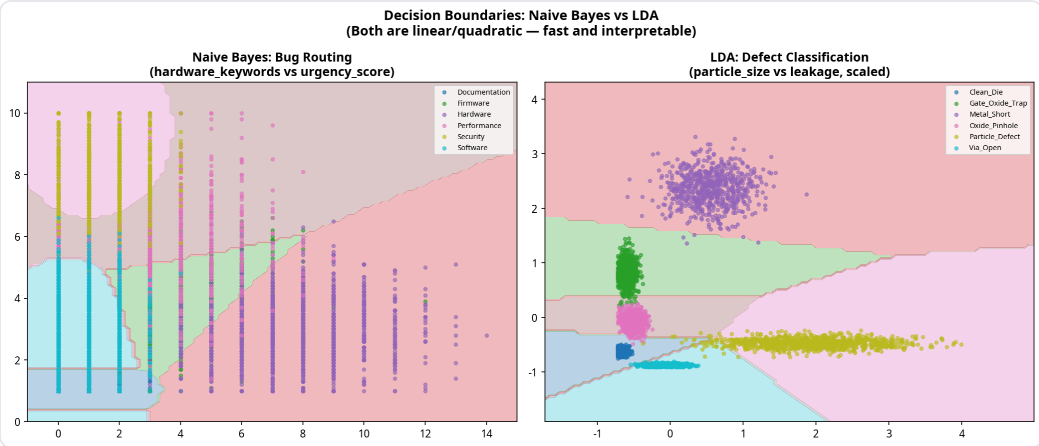
Proj1 Jira 3D Nb Probability



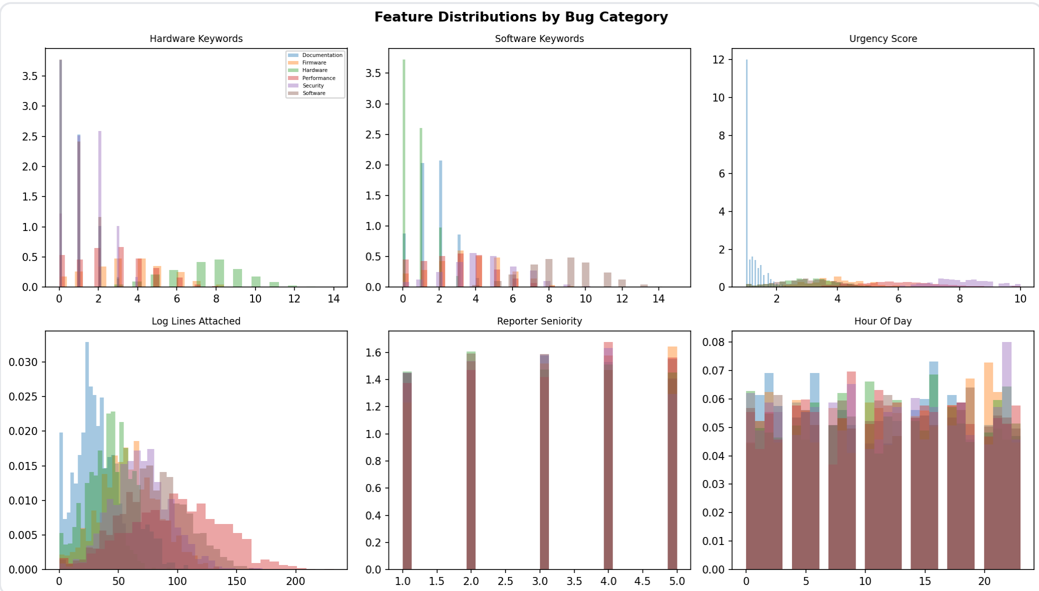
Proj1 Jira Bayes Theorem



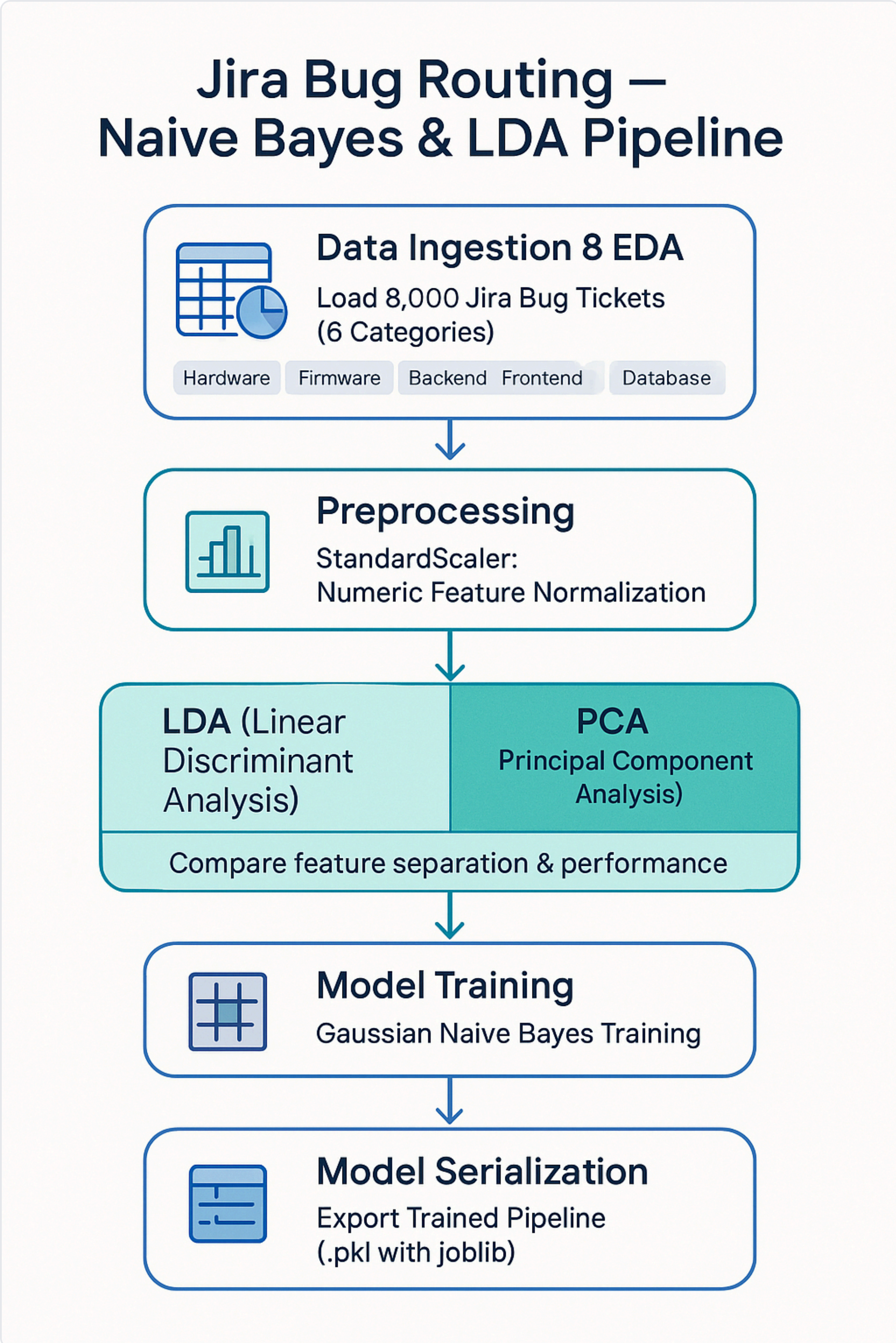
Proj1 Jira Confusion



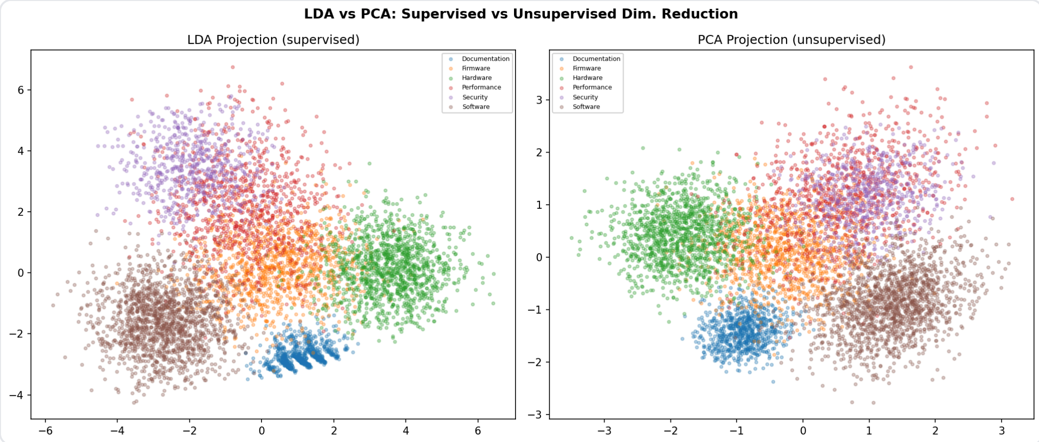
Proj1 Jira Decision Boundaries



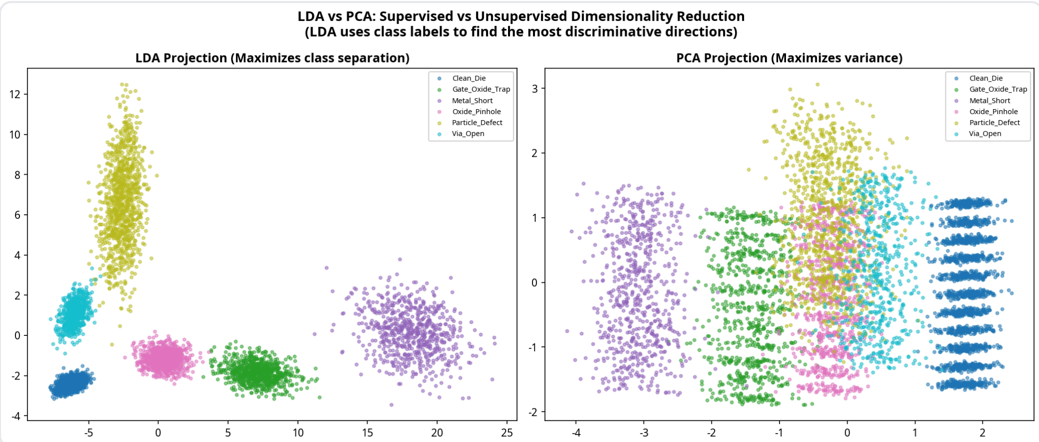
Proj1 Jira Eda



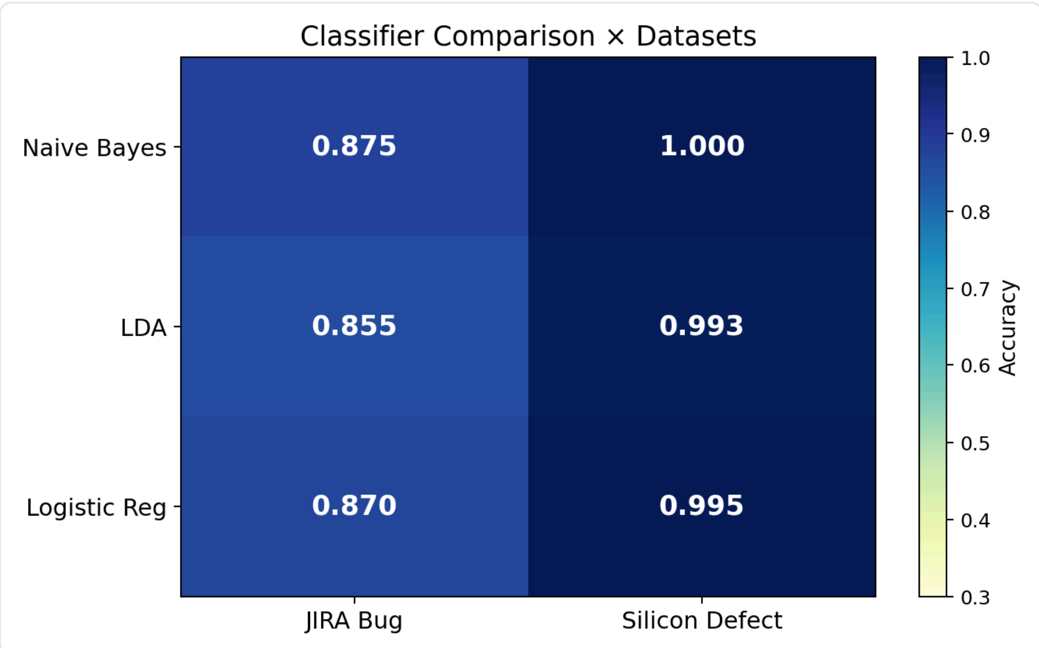
Proj1 Jira Flowchart



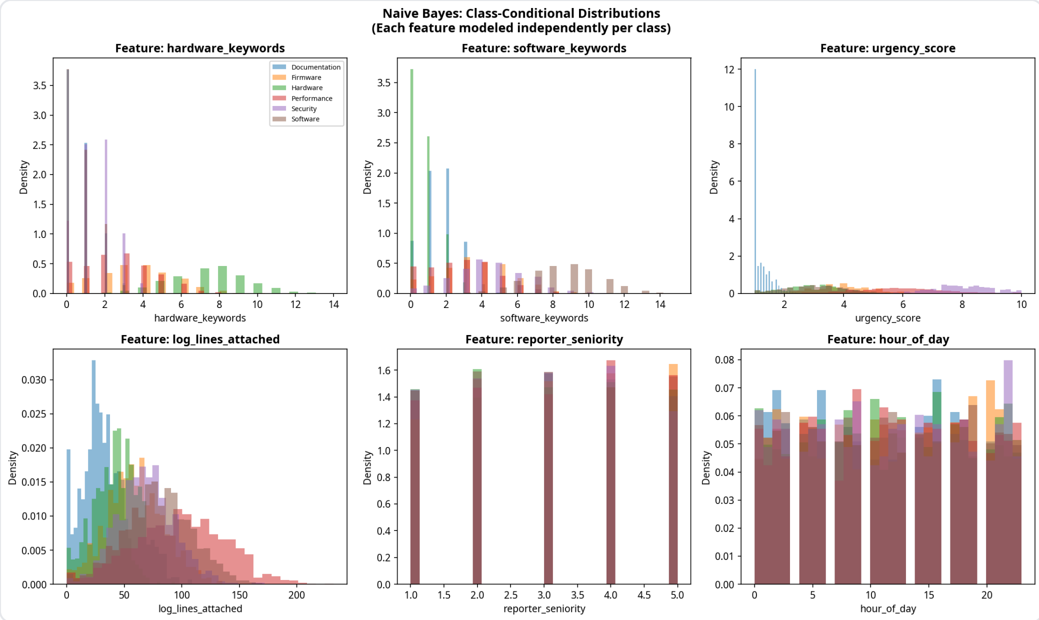
Proj1 Jira Lda Vs Pca



Proj1 Jira Lda Vs Pca Comparison

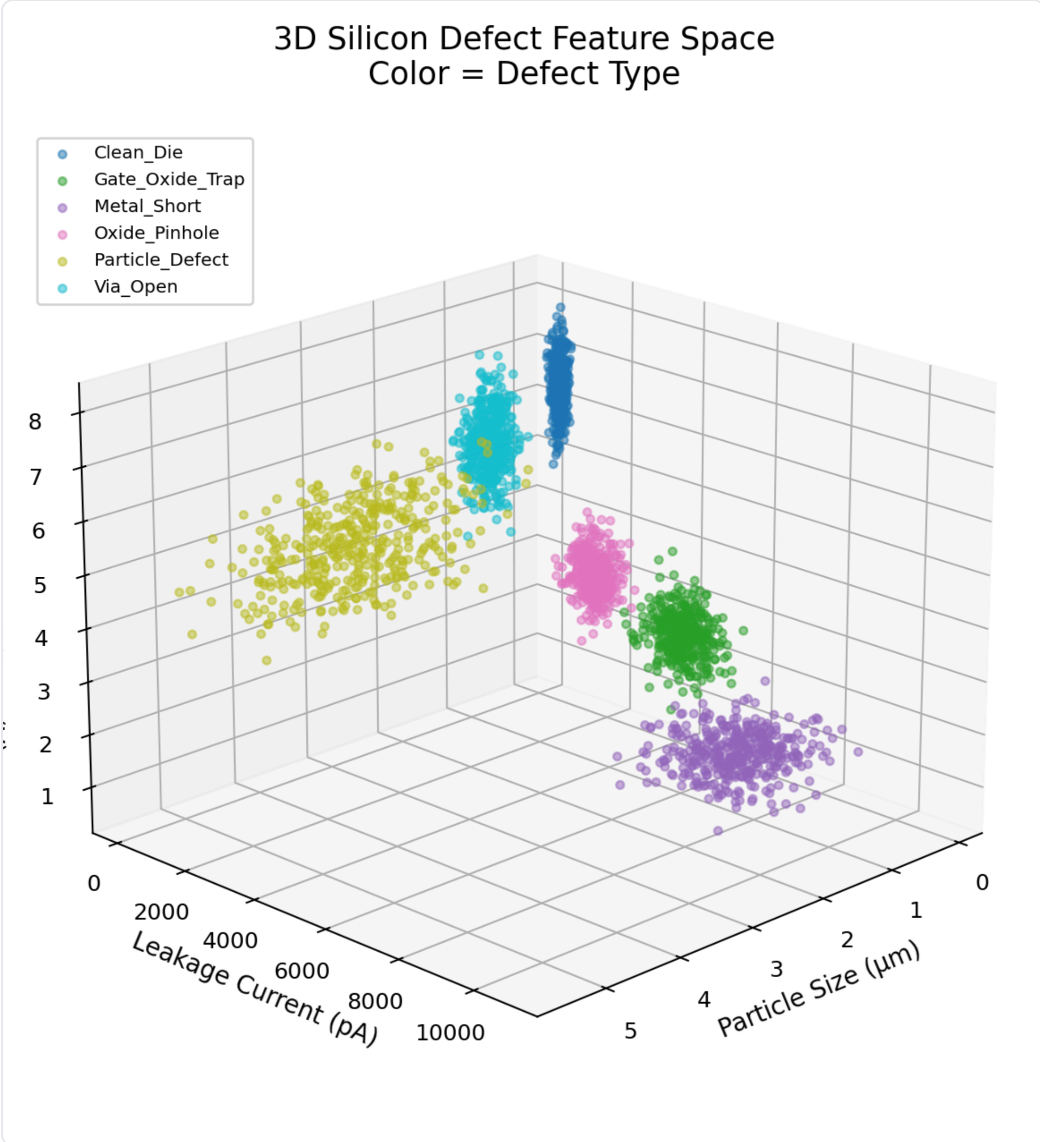


Proj1 Jira Model Heatmap

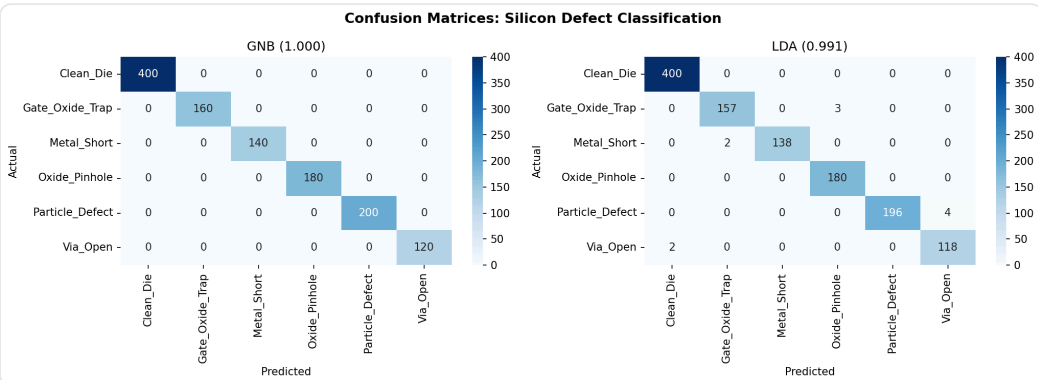


Proj1 Jira Nb Distributions

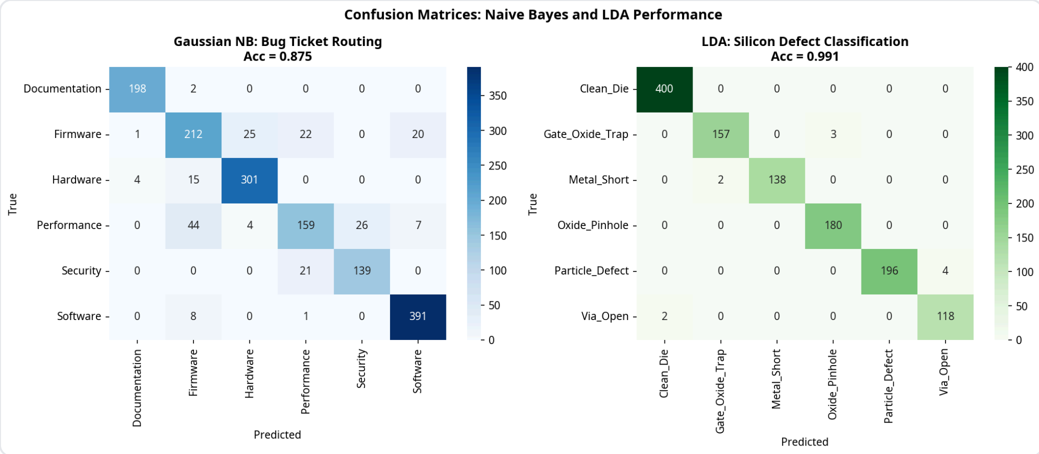
Project 2 — Silicon Defect Classification Visualizations



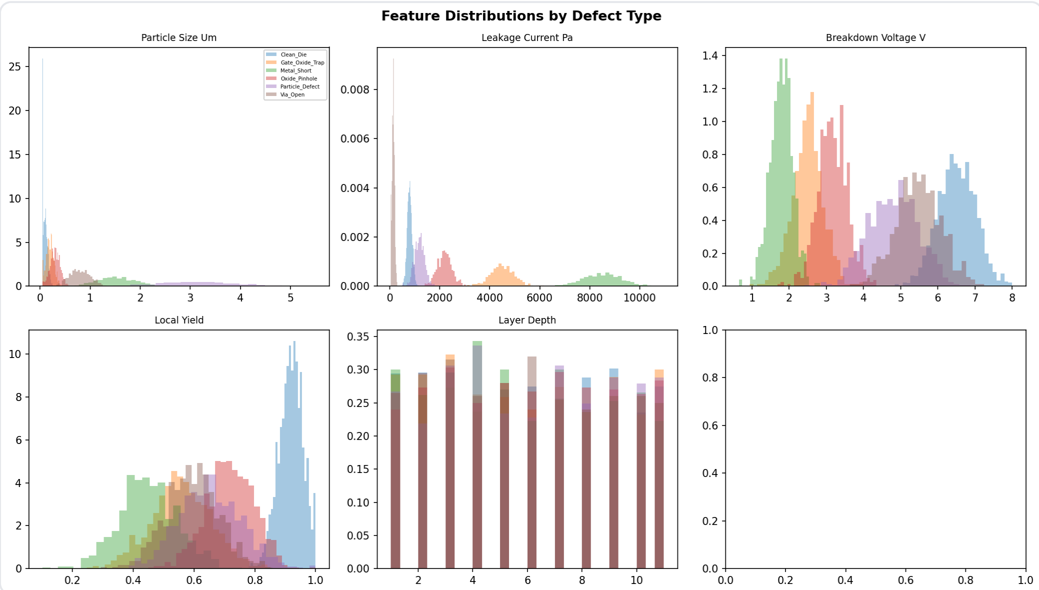
Proj2 Silicon 3D Defects



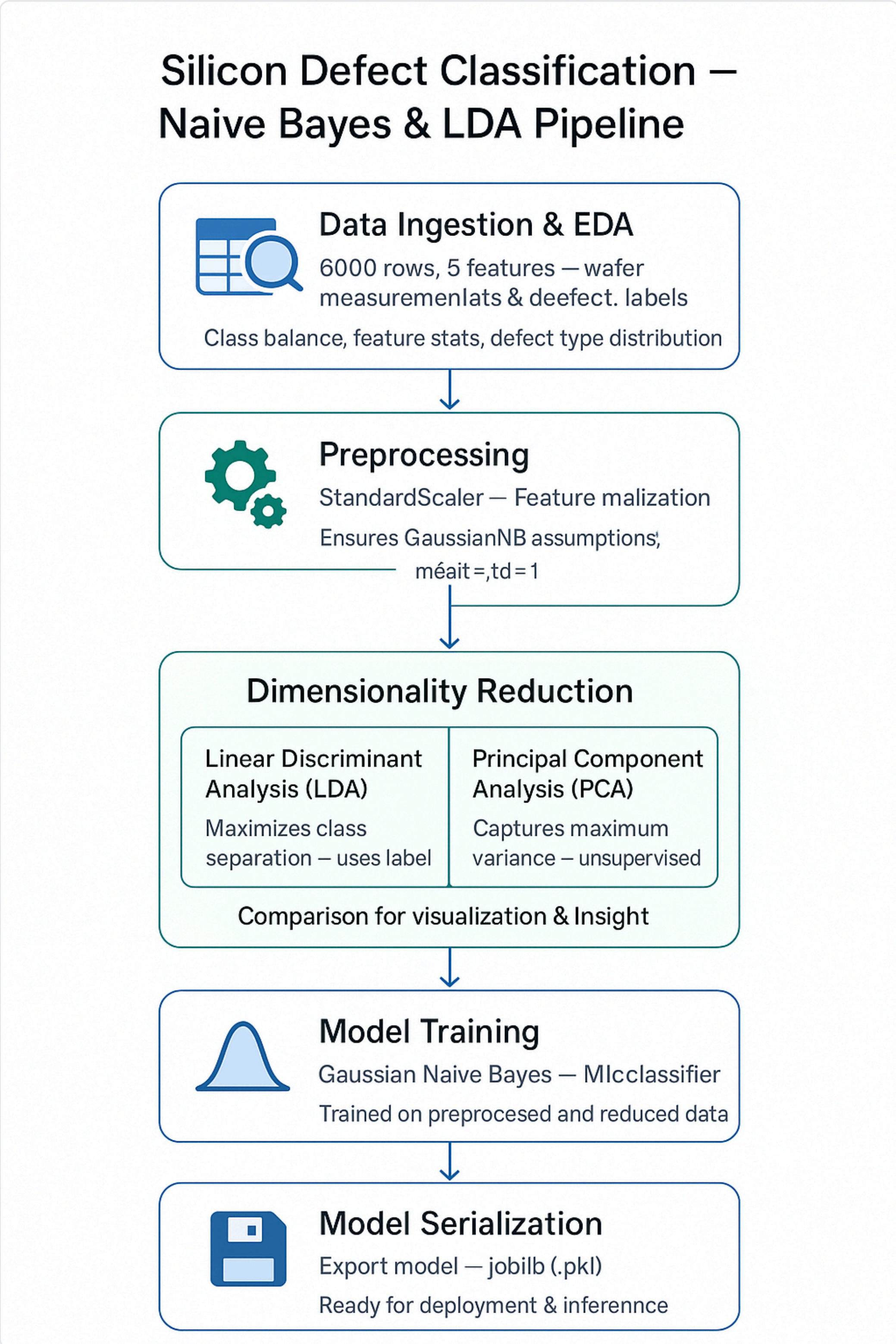
Proj2 Silicon Confusion



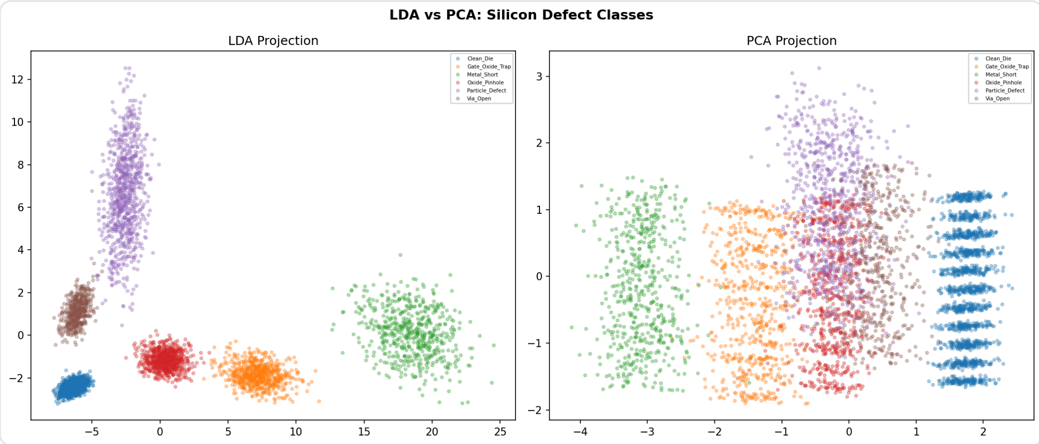
Proj2 Silicon Confusion Matrices



Proj2 Silicon Eda



Proj2 Silicon Flowchart



Proj2 Silicon Lda Vs Pca

Architecture

- **Training:** `src/train.py` — DATASETS dict pattern, trains both models
- **Inference:** `src/predict.py` — loads any model, demos both datasets
- **API:** `src/api.py` — FastAPI endpoint serving GaussianNB (Jira default)
- **Testing:** `tests/test_model.py` — 4 tests (existence + prediction × 2)